ICAD RT-5 · Set 6 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

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Numerical — write the value

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Chemistry

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Numerical — write the value

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Mathematics

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Numerical — write the value

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Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 6

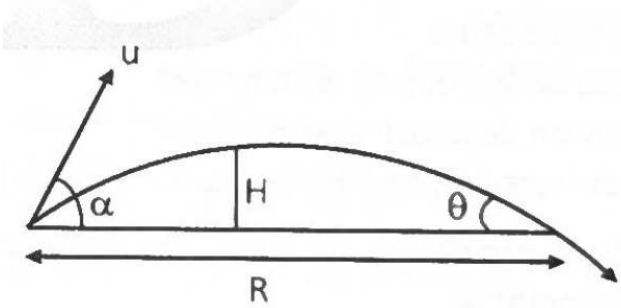
Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

Physics

PART A · Q1–20 · SINGLE CORRECT

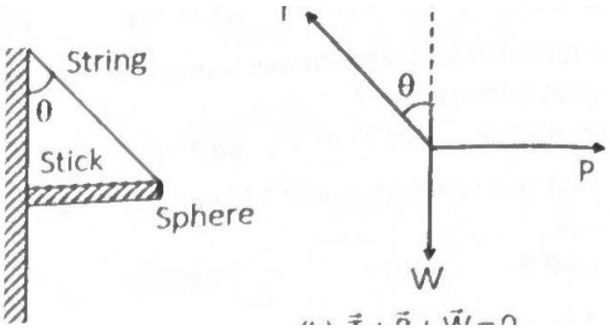
1. P4L4V5Q9 A stone is projected upward with a velocity of 20 ms^{-1}. The air resistance causes a deceleration of 2 ms^{-2} (constant, opposing the motion; take g = 10 ms^{-2}). Then (one or more correct), (a) the maximum height attained by the stone is 16.67 m (b) the time of flight of stone is 3.7 s (c) the speed with which it will strike the ground is zero (d) the acceleration during down motion is 8 ms^{-2}

2. P4L7VSuggQ5 Fig. shows the projectile motion (projected with speed u at angle α, max height H, range R, striking the ground at angle θ with speed v) in the absence of air resistance. If air resistance is taken into account, then (one or more correct): (a) time taken by projectile during upward motion is lesser than time taken during downward motion (b) θ > α (c) the value of height attained and of range of projectile decreases (d) v < u

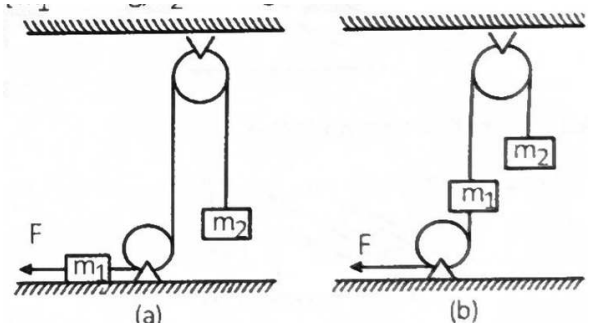


3. P4SBVPYQQ7 A paratrooper jumps from an aeroplane and opens a parachute after 2 s of free fall and starts decelerating with 3 m/s^2. At 10 m height from ground, while descending with the help of parachute, the speed of paratrooper is 5 m/s. The initial height of the aeroplane is _____ m. (g = 10 m/s^2) (a) 62.5 (b) 92.5 (c) 20 (d) 82.5

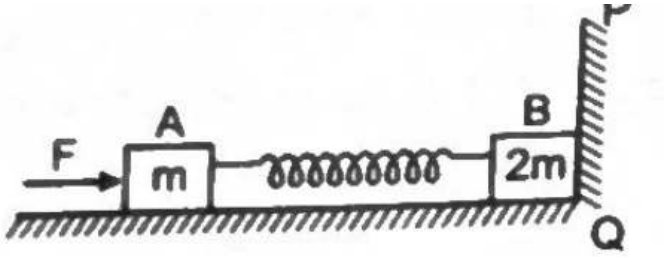
4. P5L3V5Q9 A metal sphere is hung by a string fixed to a wall. The sphere is pushed away from the wall by a stick. The forces acting on the sphere are shown in the second diagram (tension T along the string at angle θ to the vertical, horizontal push P, weight W). Which of the following statements is wrong? (a) P = W tanθ (b) $\bar{T} + \bar{P} + \bar{W} = 0$ (c) $T^2 = P^2 + W^2$ (d) $\bar{T} = \bar{P} + \bar{W}$



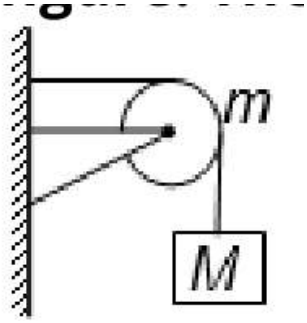
5. P6L4V5Q10 The ratio of tensions in the string connected to the block of mass m_2 in figure-(a) and figure-(b) respectively is (friction is absent everywhere) [m_1 = 50 kg, m_2 = 80 kg and F = 1000 N; take g = 10 m/s^2]. In (a) F pulls m_1 horizontally along the floor and m_1's string passes round a pulley on the floor, up over a ceiling pulley, down to m_2. In (b) F pulls the free end of a string that passes round the floor pulley up to m_1, and m_1 is tied by a string over the ceiling pulley to m_2. (a) 7:2 (b) 2:7 (c) 3:4 (d) 4:3



6. P6L5V6Q6 The figure shows a system of two blocks connected by a spring, kept on a frictionless surface: a horizontal force F pushes block A (mass m) towards block B (mass 2m), which rests against a vertical wall PQ. Block A is in equilibrium. Suddenly the supporting vertical wall PQ is removed. (a) Initial acceleration of block A is F/m (b) Initial acceleration of block B is F/2m (c) Initial acceleration of block B is zero (d) Initial acceleration of block A is F/3m



7. P6XVRT4Q16 A string of negligible mass going over a clamped pulley of mass m supports a block of M as shown in the figure. The force on the pulley by the clamp is given by (a) √2 Mg (b) √2 mg (c) $\sqrt{((M+m)^2+m^2)}\,g$ (d) $\sqrt{((M+m)^2+M^2)}\,g$



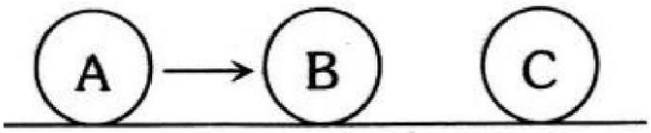
8. P7L3V3Q3 A particle is projected along the line of greatest slope up a rough plane inclined at 45° with the horizontal. If the coefficient of friction is 1/2, its retardation is: (a) g/(2√2) (b) g/√2 (c) (g/√2)(1 − 1/2) (d) (g/√2)(1 + 1/2)

9. P8L4V6Q2 A uniform chain of length 2 m is kept on a table such that a length of 60 cm hangs from the edge of the table. The total mass of the chain is 4 kg. What is the work done in pulling the entire chain on the table? (Take g = 10 m/s²) (a) 7.2 J (b) 3.6 J (c) 120 J (d) 1200 J

10. P9L4V5Q10 A rubber ball is released from a height of 5 m above the floor. It bounces back repeatedly, always rising to 81/100 of the height through which it falls. Find the average speed of the ball. (Take g = 10 ms -2) (a) 3.0 ms -1 (b) 3.5 ms -1 (c) 2.0 ms -1 (d) 2.5 ms -1

11. P9L4V6Q2 Consider two blocks A and B of masses m 1 = 10 kg and m 2 = 5 kg that are placed on a frictionless table. The block A moves with a constant speed v = 3 m/s towards the block B kept at rest. A spring with spring constant k = 3000 N/m is attached with the block B as shown in the figure. After the collision, suppose that the blocks A and B, along with the spring in constant compression state, move together, then the compression in the spring is, (Neglect the mass of the spring) (a) 0.4 m (b) 0.2 m (c) 0.1 m (d) 0.3 m

12. P9XVCAT6Q2 Three identical spherical balls A, B and C are placed on a table as shown in the figure along a straight line. B and C are at rest initially. The ball A hits B head on with a speed of 10 m s^{-1}. Then after all collisions (assumed to be elastic) A and B are brought to rest and C takes off with a velocity of : (a) 5 m s^{-1} (b) 10 m s^{-1} (c) 2.5 m s^{-1} (d) 7.5 m s^{-1}



13. P10L2V4Q3 If a r and a t represents radial and tangential acceleration, the motion of a particle will be circular if: (i) a r = 0 and a t = 0 (ii) a r = 0 but a t ≠ 0 (iii) a r ≠ 0 but a t = 0 (iv) a r ≠ 0 and a t ≠ 0. (a) (i), (iii) (b) (ii), (iii) (c) (iii), (iv) (d) (ii), (iv)

14. P10L5V4Q9 A pendulum of mass m, length l, has angular amplitude θ. Tension in the string at extreme position is T 1 and at bottom is T 2 . If maximum tension which the string can bear is twice the weight of bob, then the maximum value of θ is (a) cos −1 (1/3) (b) cos −1 (2/3) (c) cos −1 (3/4) (d) π/3

15. P10L5VSuggQ4 A pendulum of mass m and string length l oscillates in a vertical plane. The breaking tension of the string is 3mg. If the pendulum is released from rest at an angle θ from the vertical, the maximum possible angular amplitude without breaking the string is: (a) 30° (b) 45° (c) 60° (d) 90°

16. P10SBVPYQQ20 A disc with a flat small bottom beaker placed on it at a distance R from its centre is revolving about an axis passing through the centre and perpendicular to its plane with an angular velocity ω. The coefficient of static friction between the bottom of the beaker and the surface of the disc is μ. The beaker will revolve with the disc if: (a) $R \le \mu g/2\omega^2$ (b) $R \le \mu g/\omega^2$ (c) $R \ge \mu g/2\omega^2$ (d) $R \ge \mu g/\omega^2$

17. P11L1V1Q5 The position of the centre of mass of a cube of uniform density will be at: (a) edge of a cube (b) the centre of one face (c) the centre of the intersection of diagonals of one face (d) the geometric centre of the cube

18. P11L1V6Q3 Three particles of masses 60 g, 120 g, and 180 g are placed at the vertices of an equilateral triangle of side 1 m. Place 60 g at origin O(0, 0), place 120 g at (1, 0), place 180 g at the top vertex. Find the coordinates of the centre of mass. (a) (1/2, √3/6) (b) (7/12, √3/4) (c) (5/12, √3/3) (d) (2/3, √3/5)

19. P11L4V1Q3 A man of mass m starts moving on a plank of mass M with constant velocity v with respect to plank. If the plank lies on a smooth horizontal surface, then velocity of plank with respect to ground is (a) Mv/(m + M) (b) mv/M (c) Mv/m (d) mv/(m + M)

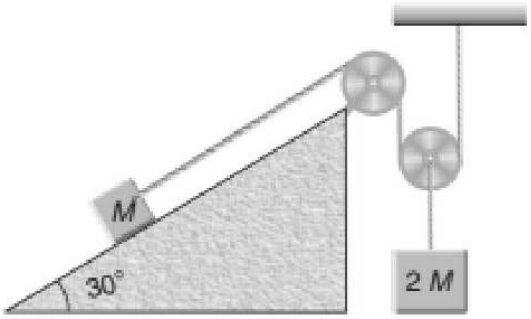
20. P11L4V5Q1 Two particles A and B initially at rest move towards each other under a mutual force of attraction. At the instant when velocity of A is v and that of B is 2v, the velocity of centre of mass of the system (a) v (b) 2v (c) 3v (d) zero

PART B · Q21–25 · NUMERICAL VALUE

21. P3XVRT4Q24 Two vectors A and B have magnitudes 2 and 2√2 respectively. It is found that $A \cdot B = |A \times B|$, then find value of $|(A + B)/(A - B)| = \sqrt{x}$, then 'x' is

22. P4XVCAT3Q24 A projectile is thrown with velocity v making an angle θ with the horizontal. It just crosses the top of two poles, each of height h, after 1 second and 3 second respectively. The time of flight (in sec) of the projectile is

23. P6XVRT3Q21 The acceleration of the block of mass M in the situation shown in the figure is g/x, then x is _____. All the surfaces are frictionless.



24. P7XVCAT4Q21 A driver's "safe speed" is the speed at which he can just brake to a halt in 30 m. On a dry road his safe speed is 20 m/s. It starts to snow and the coefficient of friction drops to half its dry value. What is his safe speed now (m/s, nearest integer)? (Neglect reaction time.)

25. P8SBVPYQQ12 If the maximum load carried by an elevator is 1400 kg (600 kg – passengers + 800 kg – elevator), which is moving up with a uniform speed of 3 m s^{-1}, and the frictional force acting on it is 2000 N, then the maximum power used by the motor is _____ kW (g = 10 m s^{-2}) — (PYQ)

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L3V6Q5 An L.P.G. cylinder weighs 14.8 kg when empty. When full, it weighs 29.0 kg and shows a pressure of 2.5 atm. During use at 27 °C, the weight of the cylinder reduces to 23.2 kg. Find the volume of n-butane in cubic metres used up at 27 °C and 1 atm. (Molecular mass of butane = 58 g mol^{-1}) (a) 1.22 m^3 (b) 1.89 m^3 (c) 2.46 m^3 (d) 3.00 m^3

2. C2L7V5Q6 In the Balmer series of lines of the hydrogen spectrum, the third line from the red end corresponds to which one of the following inter-orbit jumps of the electron for Bohr orbits in an atom of hydrogen? (a) 5→2 (b) 4→1 (c) 2→5 (d) 3→2

3. C2L7VSuggQ5 For Li^{2+} (Z = 3), find the exact wavelength (expressed as a fraction of 1/R) of the transition 3→1. Use the wavenumber formula: $\bar{\nu} = RZ^2(1/n_f^2 - 1/n_i^2)$. (a) 1/(8R) (b) 1/(5R) (c) 1/(4R) (d) 1/(9R)

4. C3L5V4Q10 How many moles of magnesium phosphate, $Mg_3(PO_4)_2$ will contain 0.25 mole of oxygen atoms? (a) 3.125×10^{-2} (b) 1.25×10^{-2} (c) 2.5×10^{-2} (d) 0.02

5. C3L12V5Q6 Carbon and oxygen form two compounds. Carbon content in one of them is 42.9% while in the other it is 27.3%. The data is in support with (a) law of definite proportions (b) law of reciprocal proportions (c) law of multiple proportions (d) law of conservation of mass

6. C4L2V5Q9 In a heteronuclear molecule A–B with a very small electronegativity difference (so that r AB ≈ r A + r B), the bond length is 1.40 Å and the covalent radius of A is 0.65 Å. Calculate the covalent radius of B. (a) 0.55 Å (b) 0.65 Å (c) 0.75 Å (d) 0.85 Å

7. C4L4V4Q5 The maximum tendency to form unipositive ion stands for the element with the electronic configuration: (a) $1s^2, 2s^22p^6, 3s^2$ (b) $1s^2, 2s^22p^6, 3s^1$ (c) $1s^2, 2s^22p^6, 3s^23p^2$ (d) $1s^2, 2s^22p^6, 3s^23p^3$

8. C4L5V5Q2 Which element will have the greatest negative electron gain enthalpy? (a) $[Ne]3s^23p^3$ (b) $[Ne]3s^23p^4$ (c) $[Ne]3s^23p^5$ (d) $[Ne]3s^23p^64s^13d^5$

9. C5L2V5Q6 NH_3 is added to BF_3 by: (a) ionic bond (b) covalent bond (c) dative bond (d) molecular bond

10. C5L2VSuggQ2 The formal charge on carbon in CO_3^{2-} is: (a) +2 (b) +1 (c) 0 (d) −1

11. C5L9VSuggQ3 For the molecules H_2O, $BeCl_2$, SO_2, and BF_3, identify all molecules that have non-zero dipole moment. (a) H_2O only (b) H_2O and SO_2 (c) H_2O and $BeCl_2$ (d) SO_2 and BF_3

12. C5SBVPYQQ18 Given below are two statements: Statement (I) : Experimentally determined oxygen–oxygen bond lengths in O_3 are found to be the same, and the bond length is greater than that of an O=O double bond but less than that of a single O—O bond. Statement (II) : The strong lone pair–lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond (O=O) but more than that of a single bond O—O. (a) Both Statement I and Statement II are false (b) Statement I is false but Statement II is true (c) Statement I is true but Statement II is false (d) Both Statement I and Statement II are true

13. C5XVCAT5Q42 A molecule with the formula AX_4Y has all its elements from p-block. Element A is rarest, monoatomic, non-radioactive from its group and has the lowest ionization enthalpy value among A, X and Y. Elements X and Y have first and second highest electronegativity values respectively among all the known elements. The shape of the molecule is (a) pentagonal planar (b) octahedral (c) square pyramidal (d) trigonal bipyramidal

14. C6L3VSuggQ2 An ideal gas undergoing expansion in vacuum shows: (a) ΔU = 0 (b) W = 0 (c) q = 0 (d) all of these

15. C6L5V3Q1 A gas expands from 1.2 m 3 to 2.4 m 3 against a pressure of 100 kPa. Convert the result to joules and then to kJ. (a) 120 kJ (b) 100 kJ (c) 140 kJ (d) 240 kJ

16. C6L6V4Q5 ΔH for CaCO 3 (s) → CaO(s) + CO 2 (g) is 176 kJ mol −1 at 1240 K. The ΔU for the change is equal to: (a) 160 kJ (b) 165.6 kJ (c) 186.3 kJ (d) 180.0 kJ

17. C6L7VSuggQ1 When a solid melts, there is: (a) an increase in enthalpy (b) no change in enthalpy (c) a decrease in enthalpy (d) a decrease in internal energy

18. C6L8V4Q7 The heat of formation of methane, $C(s) + 2H_2(g) \rightarrow CH_4(g)$, at constant pressure is ΔH = +18500 cal at 25°C. Taking R = 2 cal K^{-1} mol^{-1}, the heat of reaction at constant volume would be: (a) 19096 cal (b) 18798 cal (c) 18202 cal (d) 17904 cal

19. C6SBVPYQQ8 A cup of water at 5°C (system) is placed in a microwave oven and the oven is turned on for one minute during which the water begins to boil. Which of the following option is true? (a) q = +ve, w = 0, ΔU = −ve (b) q = +ve, w = −ve, ΔU = +ve (c) q = −ve, w = −ve, ΔU = −ve (d) q = +ve, w = −ve, ΔU = −ve

20. C6SBVPYQQ16 A liquid when kept inside a thermally insulated closed vessel at 25°C was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters? (a) ΔU < 0, q = 0, w > 0 (b) ΔU = 0, q = 0, w = 0 (c) ΔU > 0, q = 0, w > 0 (d) ΔU = 0, q < 0, w > 0

PART B · Q21–25 · NUMERICAL VALUE

21. C1XVRT4Q47 16 g of SO x gas occupies 5.6 L at 1 atm and 273 K. What will be the value of x ?

22. C2XVCAT3Q47 The wavelength (in nm) of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be (Rydberg constant = 1.097×10^7 m^{-1}) (Round off your answer to 2 digit integer)

23. C3XVCAT4Q33 What volume of water must be added to 100 cm^3 of 0.5 M NaOH to make it 0.1 M? (A) 200 cm^3 (B) 400 cm^3 (C) 500 cm^3 (D) 100 cm^3

24. C4XVCAT4Q48 Calculate the amount of energy (kJ) required to convert 5 moles of sodium atoms in the gaseous state to form sodium ions. Ionization energy of Na = 5.1 eV/atom. (Use 1 eV = 1.602×10^{-19} J, N_A = 6.02×10^{23} mol^{-1}; round off to the nearest integer)

25. C6XVCAT6Q28 Standard entropies of X 2 , Y 2 and XY 3 are 60, 40 and 50 J K −1 mol −1 respectively. For the reaction (1/2)X 2 + (3/2)Y 2 ⇌ XY 3 ; ΔH = −30 kJ to be at equilibrium, then temperature should be: (A) 750 K (B) 1000 K (C) 1250 K (D) 500 K

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M2L2VSuggQ5 Let n(U) = 100, n(A) = 60, n(B) = 50, n(A ∩ B) = 20. Find the number of elements in (A′ ∩ B). (a) 30 (b) 50 (c) 20 (d) 80

2. M3SBVPYQQ1 The sum of all the elements in the range of f(x) = Sgn(sin x) + Sgn(cos x) + Sgn(tan x) + Sgn(cot x), x ≠ nπ/2, n ∈ Z, where Sgn(t) = 1 if t > 0 and −1 if t < 0, is : (a) 2 (b) 0 (c) −2 (d) 4

3. M4L2V4Q9 Let f(x) = { 2x + 3 ; x ≤ 1 ; a^2x + 1 ; x > 1 }. If the range of f(x) = R (set of real numbers), then the number of integral value(s) which a may take: (a) 2 (b) 3 (c) 4 (d) 5

4. M5ASBVPYQQ1 If cot x = 5/12 for some x ∈ (π, 3π/2), then sin 7x (cos(13x/2) + sin(13x/2)) + cos 7x (cos(13x/2) − sin(13x/2)) is equal to (a) 6/√26 (b) 4/√26 (c) 1/√13 (d) 5/√13

5. M5BXVCAT3Q59 If tan α = x/(x + 1) and tan β = 1/(2x + 1) (α, β acute, x > 0), then α + β is equal to (a) π/2 (b) π/3 (c) π/6 (d) π/4

6. M5CL1V5Q2 The value of $\sin^2(\pi/18) + \sin^2(\pi/9) + \sin^2(7\pi/18) + \sin^2(4\pi/9)$ is: (a) 1 (b) 2 (c) 4 (d) none

7. M6L2V3Q4 Determine the solution set of x > −2 and x ≤ 3. (a) (−2, 3] (b) (−2, 3) (c) [−2, 3] (d) [−2, 3)

8. M6L3V5Q7 If $|x^2 - x - 6| = x + 2$, then the number of solutions will be: (a) 0 (b) 1 (c) 2 (d) 3

9. M6XVCAT4Q64 If $(x - 1)(3 - x)(x - 2)^2 > 0$, then x belong to (a) (1, 2) ∪ (2, 3) (b) (−1, 2) ∪ (2, 3) (c) (1, 0) ∪ (2, 3) (d) none of these

10. M7L1VSuggQ4 If a^x = b^y and a, b > 0, which of the following is necessarily true? (a) log a^x = log b^y (b) x log a = y log b (c) Both a and b (d) Neither a nor b

11. M7XVRT4Q65 If x = log_(2a)(bcd/2), y = log_(3b)(acd/3), z = log_(4c)(abd/4) and w = log_(5d)(abc/5) and 1/(x + 1) + 1/(y + 1) + 1/(z + 1) + 1/(w + 1) = log_(abcd) N + 1, the value of N is (a) 40 (b) 80 (c) 120 (d) 160

12. M8ASBVPYQQ5 If the sum of the first four terms of an A.P. is 6 and the sum of its first six terms is 4, then the sum of its first twelve is (a) −20 (b) −22 (c) −26 (d) −24

13. M8BXVCAT5Q63 For 0 < φ < π/2, if x = Σ_(n=0)^∞ cos^(2n)φ, y = Σ_(n=0)^∞ sin^(2n)φ, z = Σ_(n=0)^∞ cos^(2n)φ sin^(2n)φ, then (a) xyz = xz + y (b) xyz = xy + z (c) xyz = x + y + z (d) Both B and C

14. M8DL1V5Q3 The sum of n terms of the following series 1 + (1 + 3) + (1 + 3 + 5) + (1 + 3 + 5 + 7) + ... (a) n^2 (b) $[n(n+1)/2]^2$ (c) n(n+1)(2n+1)/6 (d) n(n+1)/2

15. M8DL2V6Q6 The value of Σ (r = 1 to n) $(1^3 + 2^3 + 3^3 + ... + r^3)/(r + 1)$ is (a) n(n+3)(n+5)(3n+2)/48 (b) n(n−1)(n−2)(3n−1)/48 (c) n(n+1)(n+2)(3n+1)/48 (d) n(n−3)(n−5)(3n−2)/48

16. M9L2V5Q4 If 2 + i√3 is one root of $x^2 + px + q = 0$, where p, q are real, then p, q = (a) 4, 7 (b) 1, 1 (c) −4, 7 (d) 4, 4

17. M9L4VSuggQ1 The triangle inequality for complex numbers states: (a) $|z_1 + z_2| \geq |z_1| + |z_2|$ (b) $|z_1 + z_2| \leq |z_1| + |z_2|$ (c) $|z_1| + |z_2| = 0$ (d) $|z_1 + z_2| = |z_1 - z_2|$

18. M9L5V2Q4 For a > 0, arg(ia) = (a) π/2 (b) -π/2 (c) π (d) -π

19. M9L5V5Q6 If $\arg(z_1) = \pi/2$, $\arg(z_2) = \pi/4$, then $\arg(z_1 z_2^2)$ = (a) 0 (b) 3π/2 (c) π (d) None of these

20. M9L7V6Q2 The value of Σ (n=1 to 5) $(x^n + 1/x^n)^2$ where $x^2 - x + 1 = 0$ is (a) 8 (b) 10 (c) 12 (d) 16

PART B · Q21–25 · NUMERICAL VALUE

21. M4XVRT1Q60 Let f(x) be a degree-5 polynomial with leading coefficient unity such that f(1)=5, f(2)=4, f(3)=3, f(4)=2, f(5)=1. Then f(6) equals: (A) 0 (B) 24 (C) 120 (D) 720

22. M7L3V3Q2 The value of $3 \cdot \log_{10} \sqrt[3]{10}$ is _____.

23. M8AL1VSuggQ8 Let T_r be the r th term of an A.P. whose first term is a and common difference is d. If for some positive integers m, n with m ≠ n we have $T_m = 1/n$ and $T_n = 1/m$, then find the value of (a − d).

24. M8BL1V4Q2 If the 12th term of a G.P. whose 8th term is 192 and common ratio is 2 is λ·2^μ, where λ is an odd integer, then λ + μ is

25. M8DXVCAT6Q72 If Σ k=1 10 k/(k 4 + k^2 + 1) = m/n, where m and n are co-prime, then m + n is equal to